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Customizing Conversation: Participatory Design of an LMM-Powered Digital Food List with Older Adults

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Nicholas Isaac Falkman Doxander

HALMSTAD UNIVERSITY

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Nicholas Isaac Falkman Doxander

nicfal20@student.hh.se

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Abstract. *Voice assistants hold promise for supporting independent living among older adults, but traditional voice assistants rely on rigid turn-taking that fails to accommodate natural speech patterns. While recent research employs large multimodal models to enable fluid conversation, most solutions rely on automatic adaptation which risks perpetuating benevolent paternalism, inadvertently infringing upon user agency by making choices for them. Granting older adults' manual control also introduces a "double-edged sword": satisfying their psychological need for autonomy while introducing extraneous cognitive load. The present study investigates how older adults with varying sensory and physical abilities experience the tension between autonomy and cognitive burden when customizing the conversational behaviors of a voice assistant powered by a large multimodal model. Adopting Participatory Design, this study shifts older adults from passive subjects to active co-creators. Seven older adults participated in interviews and co-design sessions, using Google Stitch and the ChatGPT application as technology probes to ground abstract concepts of AI customization. Findings recast manual customization as a fundamental accessibility requirement, indicating a narrow window for demonstrating basic usability before technology abandonment. This highlights the necessity of multimodal customization enabled through both voice and a graphical user interface to grant older adults a verifiable degree of control.*

Key words: Older adults, Voice assistants, Large multimodal models, LMMs, LMM-powered voice assistants, Participatory design, Co-design

1 Introduction

Cognitive and physiological changes associated with an aging global population heavily impact the daily lives of older adults (OAs) (World Health Organization [WHO], 2025). Voice assistants (VAs) have therefore been highlighted as a viable alternative to traditional graphical interfaces by allowing for a more accessible mode of interaction with technology (Pradhan et al., 2020). Yet, traditional VAs introduce their own usability barriers. Their reliance on "speak-wait/speak-wait" turn-taking fails to support the natural speech patterns of OAs, who often pause mid-sentence to gather their thoughts (Liu et al., 2025; Pinto & Belpaeme, 2024).

The emergence of Large Language Models (LLMs) as the technology powering newer VAs however allows for more dynamic conversations for OAs, including real-time speech rate and intelligent turn-taking adjustments (Eskandari et al., 2025; Pinto & Belpaeme, 2024; Wang et al., 2023), conversational repair through "barge-in" interruptions and supportive backchanneling (Liu et al., 2025), and the delivery of empathetic, non-patronizing system personas (Guo et al., 2025). Large Multimodal Models (LMMs), the main focus of this thesis, extend these text-based LLMs further. Unlike LLMs that just process text, LMMs treat audio natively, allowing the VA to directly process tone, pauses, and nuances of human speech in real-time (Wang et al., 2025). While recent research has identified some of the conversational requirements OAs need to overcome usability challenges, it is important to note that these requirements have been established across studies investigating both "classical" rule-based VAs and newer LLM-powered VAs (Franceschetti et al., 2024; Liu et al., 2025). However, whether the underlying technology relies on the rigid turn-taking of traditional VAs or the open-ended capabilities of LLMs, current implementations often rely on the system to automatically adapt to the user in the background (Eskandari et al., 2025; Liu et al., 2025; Wang et al., 2023). This system-driven approach risks lowering user agency and borders on benevolent paternalism, where technology attempts to protect or assist OAs without their consent (Horstmann et al., 2023). Because OAs desire to maintain independence and dislike technology that feels patronizing or out of their control (Horstmann et al., 2023; Mahmood et al., 2025), this thesis challenges the notion of a system-driven approach through giving OAs the agency to customize the conversational behaviors as they relate to the identified conversational requirements.

The aim with this study is to understand OAs' preferences for customizing and controlling the behavior of LMM-powered VAs through a Participatory Design (PD) framework. To ground the exploration in concrete experiences, this research utilizes existing AI tools as technology probes. In collaboration with Alla hjärtans hus in Halmstad, participants co-design the graphical user interface (GUI) of an envisioned digital food list using Google Stitch, while simultaneously interacting with and directly customizing the conversational behaviors of the OpenAI ChatGPT voice application. The goal of deploying these probes is not to create a finalized customization interface, but rather to use them as experiential primers to facilitate dialogue. By allowing older users to immediately test what it feels like to shape the VA's behavior, this study attempts to empower OAs to weigh in on the trade-offs of having the VA's behavior decided for them automatically, and if they want to explicitly control these conversational behaviors themselves. To fulfill this goal, the thesis explores the following two research questions:

The main research question: *"How do older adults with varying sensory and physical abilities experience the tension between autonomy and cognitive burden when customizing the conversational behaviors of an LMM-powered voice assistant?"*

Secondary research question: *"In what ways does the chosen modality for customization influence their sense of agency and trust in the system?"*

2 Literature Overview

2.1 Aging and Voice Assistants

Jaul & Barron (2017) conducted a mini-review highlighting the effects of aging on individuals 85 years and older. They demonstrated that advanced aging is accompanied by age-related hearing loss, reduced vision, and adverse effects on general mobility which may lead to loss in OAs' independence. Another study by Hawthorn (2000) focused on a younger population consisting of adults from 45 to 100 years of age, but nonetheless demonstrated the negative effects on GUIs caused by age-related effects on vision, dexterity, cognition, hearing, and speech. A decline in static and dynamic visual acuity, a reduction in the ability to focus on close objects, and reduced contrast sensitivity were shown to have possible implications for interface usage as these changes may struggle with small font-sizes and low contrasts. Hawthorn (2000) also emphasises that age-related cognitive decline shows general slowing of response times and difficulties with complex motor tasks which might influence technology usage among OAs negatively. Hearing declines also rise to 75% for OAs between the ages of 75 and 79 years of age, and OAs also struggle with the ability to localize sounds and might struggle with less distinct speech patterns which could be attributed to reduced motor control of tongue and lips (Hawthorn, 2000). The multimorbidity associated with advanced aging also commonly leads to a homebound status where OAs typically need to rely on home-based services and technologies for support. (Jaul & Barron, 2017). VAs often embodied in smart speakers such as Alexa Echo, Google Home or Apple's Siri assistant have emerged as a promising technology, posited as a solution to circumvent these sensory and motor barriers and thus enabling OAs to interact with technologies in a modality that aligns better with aging populations (Horstmann et al., 2023; So et al., 2024). Unlike GUIs that demand visual acuity and fine motor dexterity, voice-based interaction is often perceived by OAs as more approachable, intuitive, and natural to use (Pradhan et al., 2020). As VAs and voice-based interactions have grown in popularity among OAs, prior research has examined their usability and perceived benefits. However, attention has also been brought to the fact that VAs present new and unique usability challenges for OAs (Horstmann et al., 2023; Kuoppamäki et al., 2023; Shandilya & Fan, 2022; So et al., 2024; Trajkova & Martin-Hammond, 2020).

2.1.1 Benefits and barriers for older adults when using traditional voice assistants

Voice user interfaces (VUIs) are often praised as being more approachable and accessible than traditional GUIs, a claim that holds particular weight for OAs with age-related physical or visual declines (Pradhan et al., 2020). VUIs have the additional benefit of lowering the barrier to technology access (Kumah-Crystal et al., 2018; Pradhan et al., 2020). Due to the ease of use, prior research has highlighted that OAs frequently prefer VUIs over traditional GUIs (Kowalski et al., 2019; Wulf et al., 2014).

Traditional VAs, due to the inherent mode of interaction, do however introduce new usability challenges. Traditional VAs often rely on rule-based dialogue (Zhong et al., 2025), which restricts

interactions and limits their flexibility when met with diverse user demands (Munz et al., 2025; Salomé et al., 2025). Unlike natural human conversations, which involve interruptions and backchannels that support dialogue (Aylett & Romeo, 2023; Skantze, 2021), traditional VAs have typically relied on "speak-wait/speak-wait" interactions and silence thresholds that often fail to adhere to the natural speech patterns of OAs who frequently pause to collect their thoughts, or sometimes speak with ambiguous expressions (Liu et al., 2025; Ziman & Walsh, 2018). Because the underlying Automatic Speech Recognition (ASR) models which traditional VAs are built upon misinterpret these natural hesitations as the end of a command, they often time out or cut older adults off mid-sentence (Ziman & Walsh, 2018). Traditional VAs have also been shown to place high cognitive demands on OAs by requiring command memorization (Guo et al., 2025; Pradhan et al., 2020), presenting multiple questions in single prompts (Mahmood et al., 2025), and answering with lengthy responses that strain short-term memory (Hawthorn, 2000; Wang et al., 2023). OAs are thus left with little room to interrupt, ask for clarification, or correct conversational breakdowns when these systems fail (Kuoppamäki et al., 2023; Liu et al., 2025; Munz et al., 2025).

2.2 The Shift to Large Multimodal Models

Large Language Models (LLMs) can potentially address the many usability challenges OAs face with traditional VAs (Guo et al., 2025; Liu et al., 2025; Mahmood et al., 2025; Munz et al., 2025). Because LLMs are built on the transformer architecture (Vaswani et al., 2017), they move away from rule-based processing and can instead capture complex meaning and context. This allows them to engage in the open-ended dialogue needed to make voice interactions more natural and accessible (Liu et al., 2025).

To alleviate the cognitive and usability challenges outlined in Section 2.1.1, LLM-powered VAs move away from rigid command-based interactions, enabling these systems to interpret and respond to more complex inputs or requests without the need for OAs to memorize specific keywords (Yang et al., 2024). VAs that embed LLMs often achieve this through a pipeline where ASR and Text-to-Speech (TTS) models are combined to enable for audio input to be captured, transcribed and processed by the LLM and then returned to the user as speech (Wang et al., 2025).

2.2.1 LLM-powered voice assistants

Driven by these capabilities, recent research on LLM-powered VAs has largely shifted focus away from basic task execution toward utilizing the VAs as emotional companions and complex information mediators. For example, recent studies have explored LLM-powered VAs designed to encourage self-disclosure and alleviate social isolation, acting as a companion for OAs to discuss stigmatized topics (Guo et al., 2025). In healthcare contexts, LLMs are being deployed to synthesize dense medical documents into accessible verbal debriefs, acting as intermediaries between homebound patients and healthcare providers (Mahmood et al., 2025; Yang et al., 2024). To make VAs feel more like empathetic companions rather than rigid machines, researchers have designed systems that allow for two-way conversations (Liu et al., 2025; Munz et al., 2025). This two-way conversation allows for co-operative interruptions; this means the VA and the user can naturally chime in to help complete a forgotten thought, supply a missing word, or ask a clarifying question, creating a much more human-like dialogue (Liu et al., 2025).
With this shift toward highly conversational and open-ended dialogue, LLM-powered VAs introduce a new "double-edged sword." Notably, the study by Munz et al. (2025), from which this term originates, evaluated participants who already possessed extensive experience using conversational agents. They highlight that the tendency of LLMs to be "verbose" can increase

cognitive load, creating a mismatch between the VA's output and the user's immediate need (Munz et al., 2025). The open-ended nature of these interactions also often creates uncertainty, leaving OAs struggling to determine how to speak or what they should ask to obtain the desired result (Kuoppamäki et al., 2023; Munz et al., 2025). To address the LLM-related usability challenges faced by OAs, previous research has proposed several conversational requirements aimed at alleviating them, which are explored in the following section.

2.3 Conversational Requirements for Older Adults

2.3.1 Conversational pace, speech rate and turn-taking

Earlier non-LLM-powered VAs have as previously stated relied on rigid interactions that have proved challenging for OAs (Liu et al., 2025). Pinto and Belpaeme (2024) addressed this limitation by exploring the use of LLMs to anticipate turn transitions. Their work demonstrated that LLM-powered VAs, by analyzing semantic meaning through text rather than simply relying on silence thresholds, can more accurately distinguish between a user pausing to reflect and a user ending their turn, a distinction that traditional VAs consistently failed to make. Beyond determining when to speak, the actual pace at which the agent delivers the information is an equally important factor. Wang et al. (2023) used a Wizard of Oz (WoZ) methodology to investigate the feedback speech rate regulation for OAs. Their study revealed that OAs often speak slower when conversing with VAs, and equally expected VAs to respond at a slower pace than their own. Wang et al. (2023) additionally found a negative correlation between the system's response word count and OAs' expectations of speech rate as verbose answers required even slower speech rate to successfully be interpreted by OAs. Building on this need for adaptive pacing, Eskandari et al. (2025) developed an LLM-powered VA able to automatically and dynamically adapt its speech rate to match the user which significantly improved the interaction with the model for the participating OAs. Taken together, these studies indicate that adapting a VA's speech rate is important for accommodating OAs' individual cognitive processing speeds in order to improve the quality of interactions. Eskandari et al. (2025) also demonstrate that automatic adaptation improves interaction quality, and serves as an example of when OAs stand to gain from automatic adaptation.

2.3.2 Conversational repair and barge-in

By enforcing the "speak-wait/speak-wait" pattern, traditional VAs restrict OAs to one-way communication, preventing them from correcting misunderstandings or asking the VA to repeat information (Liu et al., 2025). LLM-powered VAs are showing equally strong potential in alleviating these challenges as evidenced by a study by Liu et al. (2025) where the researchers built and tested an LLM-powered VA which enabled "barge-in", meaning the VA had support for both user-initiated and agent-initiated interruptions. Their study showed that allowing OAs to interrupt the VA empowered them to take control of the conversation, correct mistakes immediately, and ask for clarifications if the VA spoke too quickly. Liu et al. (2025) also highlighted the value of "co-operative interruptions" where the VA assists the user through interjections. Participants appreciated when the VA gently interjected to fill in a forgotten word or to ask clarifying questions, which kept the dialogue going and increased their ability for conversational repair.

2.3.3 Tone, persona, and avoiding patronization

Because LLMs generate open-ended responses, the personality and emotional delivery, i.e. the "persona" of the assistant is just as important. Through co-design sessions, Hu et al. (2024) found

that OAs prioritize non-functional requirements, desiring a system persona that mirrors an authentic, two-way "human" conversation without feeling overly robotic. This persona must also find a balance between sociability and respect; Horstmann et al. (2023) note that OAs have a desire to maintain their independence and strongly dislike interactions that feel patronizing or manipulative. This is especially important given that commodity LLMs often encode societal age-biases and stereotypes, which can inadvertently lead to unintended although patronizing outputs if not mitigated (Dewan et al., 2025). When designing an LLM-powered VA for OAs, Guo et al. (2025) found that OAs desire a VA that functions as "neutral third party" and a "preferred confidant" providing agreeable, empathetic responses without imposing personal beliefs. Allowing the customization of the VA's tone and persona is therefore important to avoid patronization and align the VA with OAs' preferences.

2.4 Advancing to Large Multimodal Models

While recent studies highlight the necessity of the previously documented conversational requirements for LLM-powered VAs, the actual implementation of these features in LLM-powered VAs has been hindered by technical architectures. Most papers identified in this study implement LLMs as part of a pipeline built upon several distinct models to deliver the final system. While this modular approach improves semantic understanding, it often introduces processing latency and generally (due to the text-based processing) still struggles with cues such as users' tone, emotional state, hesitation and speech rate, which are lost during the ASR transcription phase (Wang et al., 2025).

To overcome these limitations, VA development is currently shifting toward the use of Large Multimodal Models (LMMs), also referred to as omni-modal models or Auditory-LLMs (Wang et al., 2025). As opposed to relying on a pipeline as mentioned, LMMs (such as OpenAI's GPT-4o) treat audio natively. This allows the model to process audio directly and preserves the emotional nuances and information inherent in human speech.
As Wang et al. (2025) highlight, this shift to native "machine listening and speaking" enables these systems, and by extension VAs, to more effectively interact through sound. Because LMMs process audio natively rather than relying on ASR pipelines, they provide the technical foundation to potentially meet the conversational requirements that OAs have been shown to exhibit, specifically the requirements that relate to *speech rate*, *voice type*, *turn detection*, and *barge-in behavior*.

Most significantly, LMMs are capable of true "full-duplex" interaction, meaning they can "listen" and "speak" simultaneously. Wang et al. (2025) also note that full-duplex communication is an important feature for mimicking human-like interaction as it allows the model to natively process backchannel cues and perform or allow for barge-ins without the awkward delays characteristic of ASR-based systems. This makes these models ideally positioned to adapt to the slower speech patterns, longer pauses, and diverse tonal variations often exhibited by OAs.
Currently, empirical research exploring the use of LMMs in participatory and co-designing conversational experiences with OAs remains scarce, at least to the best of this thesis's author's ability to locate such work. Given that seminal LMM research only emerged in 2025 (Wang et al., 2025), it is also possible that relevant studies exist but have yet to be published, or the field is simply too nascent for such work to have been completed. As such, using an omni-modal architecture serves as a methodological choice for this study. By utilizing an LMM capable of full-duplex communication, this research can potentially bypass the technical limitations of prior ASR-based systems and evaluate how OAs experience the agency to customize the conversational behavior of

an LMM-powered VA. While LMMs make both manual customization and automatic adaptation technically achievable, the question of who controls their configuration remains open.

2.5 Automatic Adaptation and User Control

While recent research has identified the conversational requirements that solve for the usability challenges that traditional and LLM-powered VAs introduce for OAs, most of these proposed solutions rely on the system to automatically adapt to OAs (Eskandari et al., 2025; Liu et al., 2025; Wang et al., 2023). This risks removing user agency and potentially perpetuates benevolent paternalism, a phenomenon where creators of technology attempt to protect OAs, but inadvertently infringe upon their agency and make them feel patronized (Horstmann et al., 2023). Because OAs show a desire to maintain independence and dislike technology that feels patronizing or out of their control (Horstmann et al., 2023), relying primarily on automatic adaptation potentially overlooks their need for autonomy. Supporting this desire for manual control, Franceschetti et al. (2024) through their study equally identified that OAs actively want to customize VA features to fit their preferences and hearing difficulties, ideally using a supporting screen.

While granting OAs direct control over the VA's behavior satisfies their psychological need for autonomy, it also introduces a recognized trade-off. This reflects what Munz et al. (2025) describe as the "double-edged sword" of flexible LLM interfaces. Although customization empowers user agency, its open-ended nature can inadvertently introduce cognitive load and potentially overwhelm users who must figure out exactly how to instruct the system. While this study recognizes the tension between user agency and cognitive overload, current literature frequently recommends automatic adaptations of the identified conversational requirements without providing a clear rationale for why the control of these conversational requirements should be left exclusively to developers, researchers or the systems themselves. The goal of this paper is therefore to gain insight into what OAs think about this balance. By exploring their experiences and preferences, this study adds to the discussion of whether conversational requirements should be customizable by OAs, or if the cognitive burden makes it preferable to leave these adaptations up to developers, researchers, and the system.

On a final note, because traditional design processes often limit users to the role of passive testers or sources of information (Bratteteig & Wagner, 2014), evaluating true user agency requires a different approach. As a result, this thesis adopts a methodology fundamentally built on sharing decision-making power and involving OAs as equal co-creators (Bratteteig & Wagner, 2014). To facilitate this collaboration and ground often abstract AI concepts, this study uses technology probes as experiential primers (Hu et al., 2024). This shift in power dynamics, alongside the application of these probes, is explored further in the following chapter, which introduces Participatory Design as the core framework for this research and its specific application to OAs (Joshi & Bratteteig, 2016).

3 Method

3.1 Participatory Design

As noted by Salomé et al. (2025), generative AI (where LLMs and LMMs fall into) is changing both the capabilities of VAs and the methodologies needed to design them. As these technologies

grow more complex, so too does the need to make certain they are developed in ways that are functionally effective, socially meaningful, and ethically grounded. As such, this study employs Participatory Design (PD) to democratize the design process by positioning OAs as active co-creators. Mutual learning is a core concept in this approach (Bødker et al., 2010) as co-participants possess domain and context-specific knowledge that is difficult to map and understand without their involvement (Joshi & Bratteteig, 2016). Designers and researchers however hold a certain but often unspoken authority as it relates to technical understanding and decision-making (Bratteteig & Wagner, 2014). Designers and co-participants must therefore learn from each other: designers learn about co-participants' specific contexts and activities, while co-participants learn about technical possibilities (Joshi & Bratteteig, 2016). When designers and co-participants come from different communities, this collaboration serves as a valuable opportunity to identify and verify a valid design problem and challenge stereotypical assumptions, hopefully leading to more inclusive and effective design solutions (Hu et al., 2024). It is also through this mutual learning and exchange where both parties can jointly create better solutions than either party could achieve alone.

A challenge in PD is however the inherent power imbalance between technical experts and co-participants (Bratteteig & Wagner, 2014). The authors highlight the tension between the ethical commitment to share decision-making power with co-participants and the reality that designers inherently hold considerable power as technical experts. To conceptualize how this power is negotiated, they draw on the distinction between "power over" (hierarchical control) and "power to" (agency and empowerment). Because non-technical participants often lack the vocabulary to execute technical concretization, sharing power frequently requires trust, where co-participants confidently delegate technical decisions to designers to translate their ideas (Bratteteig & Wagner, 2014; Hu et al., 2024). This study seeks to navigate these complexities by using design probes as experiential primers and giving participants the necessary experience to make informed design suggestions (Hu et al., 2024) while working toward a participatory design result that increases the OAs' "power to" act, expanding their autonomy over if and how LMM-powered VAs are customized (Bratteteig & Wagner, 2014, 2016).

3.2 Participatory Design in Practice with Older Adults

3.2.1 Participant recruitment and considerations

OAs are often assumed to be a homogeneous group, which has historically led researchers and designers to treat them uniformly, often stereotyping them as frail, dependent, or averse to technology (Pradhan et al., 2020). Salomé et al. (2025) similarly found a tendency for co-design studies to bias their participant selection towards otherwise healthy and active individuals, a practice that risks ignoring the very limitations the technology is meant to address, deepening rather than reducing existing social disparities. OAs are, however, a highly diverse demographic whose needs are influenced by varying cognitive abilities, physical health, and experiences with technology (Pradhan et al., 2020; Salomé et al., 2025). As Joshi and Bratteteig (2016) state, "Old people are as different as everyone else—maybe even more." A requirement of PD is therefore to acknowledge this diversity so the design process can be flexibly adapted to leverage the strengths and accommodate the limitations of each participant (Joshi & Bratteteig, 2016; Salomé et al., 2025). While this study aims for continuous involvement, this can be achieved with varying degrees of participation depending on OAs' abilities. PD literature also acknowledges that co-participants do not need to be involved in every phase to achieve a participatory result, particularly during technical concretization where designers' expertise naturally dominates (Bratteteig & Wagner, 2014, 2016).

This is especially relevant when involving OAs, who often experience fatigue during extended sessions. It is therefore important for researchers to shoulder the burden of artefact creation and accommodate OA participants to the best of their abilities (Joshi & Bratteteig, 2016).

3.2.2 Familiar environments and community trust

Trust is a strong component influencing successful collaboration within PD with OAs. Researchers therefore strongly advocate for collaborating with local community organizations rather than inviting participants to unfamiliar settings (Joshi & Bratteteig, 2016; So et al., 2024). Conducting design activities within a familiar space also makes the research process feel less threatening and more socially acceptable as the research becomes integrated into OAs existing routines, allowing participants to observe, join on their own terms, and keep each other updated on the project. Recruiting from a community also naturally facilitates "snowballing" recruitment, where initial participants recruit their peers, a method proven to be highly effective for engaging this demographic (Joshi & Bratteteig, 2016).

3.2.3 Design probes and Immediacy

Open-ended design tasks and abstract brainstorming can be challenging for OAs, often leading to design paralysis when dealing with unfamiliar technologies such as AI (Hu et al., 2024; Joshi & Bratteteig, 2016). To address this, utilizing design probes or functional prototypes as experiential primers grounds the conversation in the concrete, allowing participants to react to actual experiences rather than relying on imagination (Hu et al., 2024). In this study, a prototype of an LMM-powered food list was developed using Google Stitch to act as a design probe. Presenting a probe helps participants understand what the technology can achieve before they are asked to modify it—a method similarly employed in other research collaborating with OAs (Hu et al., 2024; Munz et al., 2025).

In addition, PD with OAs relies on the concept of immediacy: reducing the distance between "doing" and "seeing" (Joshi & Bratteteig, 2016). When dealing with abstract, invisible technologies, the processes of selecting, concretizing, and evaluating choices often collapse into a single activity (Joshi & Bratteteig, 2016). As such, this thesis combines interviews with co-creation sessions. While these qualitative methods necessitate smaller sample sizes, they yield rich and nuanced insights during the early-stage development of LLM-powered interfaces (Munz et al., 2025), making this approach applicable to the exploration of emerging LMMs.

3.3 Literature Review Process

A narrative review was conducted following Clark et al. (2021) to account for the interdisciplinary and evolving topic of AI, LLMs, and VAs within Information Systems and Human-Computer Interaction (Snyder, 2019). Given the rapid advancement of the AI field, the inclusion criteria included peer-reviewed articles and high-impact non-peer-reviewed (indicated by citations and study relevance) articles and conference papers as to not omit recent but relevant work. Searches were conducted in Scopus and Web of Science (WoS) for publications between 2014 and 2026. The 2014 start date was selected to identify VA research predating modern LLM/LMM-powered VAs. The literature collection and filtering process was carried out as follows:

Scopus: The search resulted in 428 documents. Results were sorted by the amount of citation to prioritize high-impact articles, and 64 documents were downloaded based on title and abstract relevance.

Web of Science: The search resulted in 514 documents across the publishers and organizational categories the author deemed relevant. The resulting papers followed the same methodology of selection as the Scopus search, and 59 papers were downloaded.

The combined searches initially resulted in 123 downloaded documents. After adjusting for duplicates, removing irrelevant articles, and excluding those inaccessible due to paywalls, 72 final articles were kept and imported into Zotero. These 72 articles underwent a secondary screening of their abstracts, introductions, findings and discussions before the remaining relevant papers were selected to be read in their entirety. Papers were finally kept or discarded based on relevance and a snowball sampling method (Clark et al., 2021) was used on the remaining relevant papers to locate additional literature. During the snowballing sampling, the 2014 publication date constraint was lifted to allow for the inclusion of older, foundational papers that were highly relevant to the study, especially as it pertains to papers on the effects of aging and Participatory Design. See *Appendix A* for an overview of the literature search and the full list of keywords.

3.4 Participant Recruitment

To mitigate the common sampling bias of exclusively recruiting healthy OAs (Salomé et al., 2025), this study prioritized OAs experiencing age-related sensory declines. The inclusion criteria required participants to be at least 60 years old, a standard in research targeting OAs (Shandilya & Fan, 2022). However, to counteract a common bias in similar studies that often rely on younger and perfectly healthy OAs (Salomé et al., 2025), this research intentionally sought out participants in or near their eighties (Vines et al., 2012). Including participants who are already experiencing sensory and physical declines (Jaul & Barron, 2017) helps the design process reflect the people who need adaptable interfaces the most. Participants were excluded if they possessed severe cognitive impairments that prevented informed consent, or if they resided in full-care nursing homes where personal grocery planning was no longer a relevant daily activity. Individuals with extreme or complete sensory loss were also excluded. While these demographics can and should have access to the same technologies as everyone else, fully accommodating their needs would require resources, such as sign language interpreters or accompanying nurses, that exceed the feasible scope of this thesis. Future research is thus highly encouraged to involve these individuals to ensure LMM-powered VAs are universally inclusive.
Participants were recruited through a partnership with Alla Hjärtans Hus, a local community meeting place for the elderly in Halmstad (Sweden), which allowed the research to be grounded in a familiar and comfortable environment, something that is important for building mutual trust between the researcher and participants (Joshi & Bratteteig, 2016). Recruitment began through conversations with visitors, after which a snowball sampling strategy was used to reach additional participants after co-design sessions. Recruitment through the researcher's personal network was also used, given participants followed the same inclusion and exclusion criteria.

3.5 Participants

A total of 7 participants were included in the study. A few of the participants had experiences with traditional VAs and LLMs through text-based chatting, although none of the participants had

experiences with LLM or LMM-powered VAs prior to joining the study. *Table 1* contains additional information about the participants.

Participant ID	Age	Gender	Living Situation	Prior AI Experience	Age-Related Variations	Session Length
P1	88	Male	Independently at home	No prior VA or general AI experience	Severe hearing impairment with hearing aids	~50 min
P2	78	Female	Independently with partner	Some experience with Apple Siri, and ChatGPT through chatting with it	Severe hearing impairment with cochlear implants	~1 h 7 min
P3	72	Male	Independently with partner	No prior VA or general AI experience	None self-identified	~50 min
P4	69	Female	Independently with partner	Uses Apple Siri for information searching	None self-identified	~43 min
P5	87	Male	Home-assistance	No prior VA or general AI experience	Severe hearing impairment with hearing aids	~32 min + ~1h informal conversation
P6	85	Male	Independently at home	No prior VA or general AI experience	Reduced manual dexterity (several broken fingers), moderate hearing impairment with hearing aids	~1 h
P7	84	Female	Independently at home	Experienced ChatGPT through chatting with it, but has never used VAs	None self-identified.	~1 h 3 min

Table 1. Overview of participants in the study.

3.6 Technology Probes

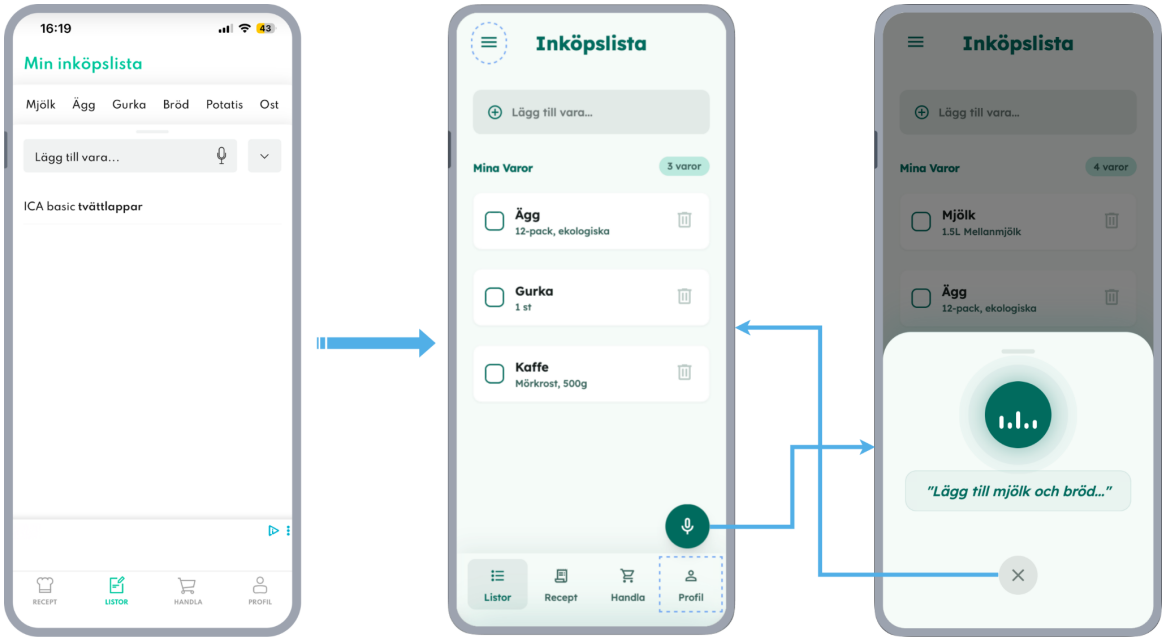
Two probes facilitated the discussions during the co-design sessions. A visual representation for a mobile application with an envisioned integrated LMM and settings for configuring the VA's conversational behavior continually developed and revised through Google Stitch, together with OpenAI's mobile application in Voice Mode. The services and subsequent technologies powering these are outlined below.

3.6.1 Google stitch

Google Stitch is (at the time of writing) a new service from Google described as an "AI-native software design canvas" (Banks, 2026), letting users generate high-fidelity, functional prototypes

from prompts and images. Changes can be made through an agent, targeting either entire views or smaller sections of the interface, while manual changes are primarily limited to text editing. This study hypothesised that the inclusion of Google Stitch would be beneficial for PD with OAs by countering the design paralysis OAs often experience when faced with the open-ended and abstract nature of traditional co-design (Hu et al., 2024). Specifically, Google Stitch was hypothesised to alleviate this paralysis through its ease of use and rapid speed of iteration by facilitating immediacy and reducing the distance between "doing" and "seeing," directly addressing the recurring challenge in PD where participants typically must wait for technical experts to interpret and execute their suggestions (Bratteteig & Wagner, 2014; Joshi & Bratteteig, 2016).

To prepare the probe, the interface was generated by providing Google Stitch with screenshots of the existing Matlistan application and a system prompt describing the study's goals and relevant parameters from the OpenAI Realtime API documentation (see Figure 1, Screen 1 & 2). Matlistan was chosen as the basis since the research was conducted in collaboration with the company Alacart, which owns the application. The researcher then adjusted the UI to fix obvious usability flaws, minimizing extraneous cognitive load so participants could focus on co-designing the customization settings instead of struggling with a cumbersome interface (Munz et al., 2025).



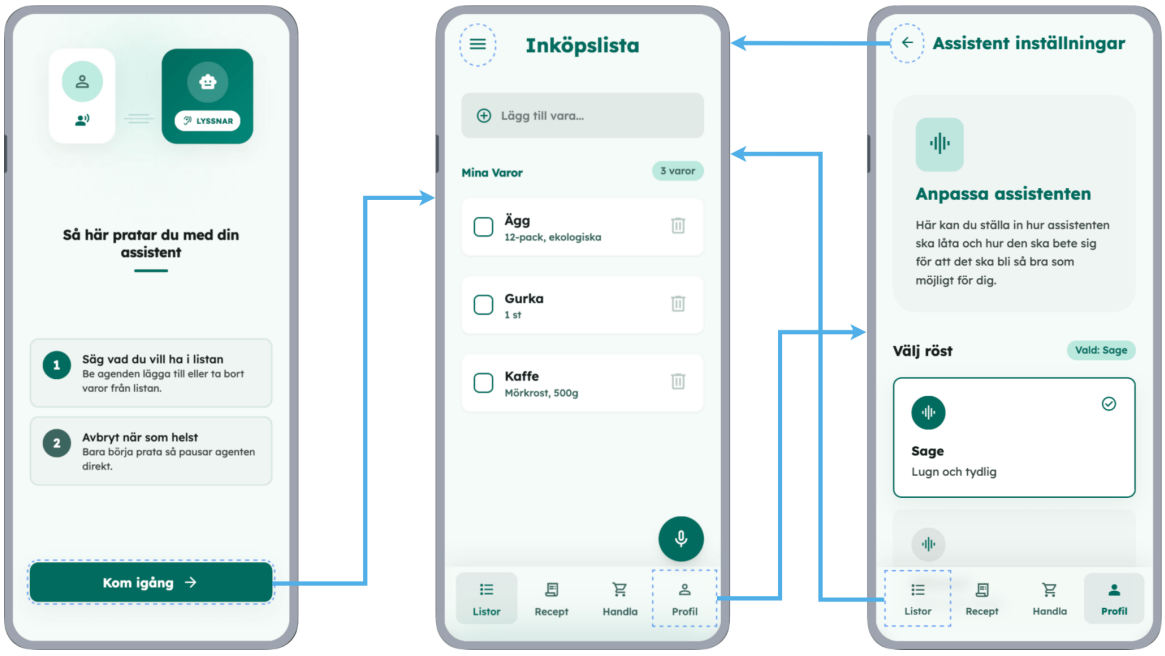
Screen 1: The existing Matlistan food list application which served as the baseline reference before the reworked prototype in Google Stitch.

Screen 2: The redesigned food list view, adapted to keep the layout similar while integrating the new voice assistant.

Screen 3 - Microphone Overlay: Shows the active listening state. While non-functional, it was used to demonstrate the envisioned interaction flow before participants tested the ChatGPT probe.

Figure 1: The baseline Matlistan (Screen 1) and the adapted Google Stitch probe (Screen 2). (Screen 3) shows the view when the envisioned LMM-powered VA is active.

To keep the probe manageable during shorter co-design sessions, the interface was limited to three views. The configuration menu (see Figure 2, Screen 6) was kept to a single screen, adhering to guidelines for simplified interactions and to reduce working memory demands (Hawthorn, 2000; Salomé et al., 2025). The configuration menu was also the focus of the co-design efforts to keep the suggestions and discussion focused.



Screen 4 - Onboarding: Explains the interactions with the voice assistant. It served as a visual guide to prepare users for how to interact with the ChatGPT probe as the Google Stitch probe did not contain this functionality.

Screen 5 - Listor: The food list view, adapted from the original Matlistan application.

Screen 6 - Profil (Configuration Menu): A single-page configuration menu to customize the envisioned voice assistant. This is where most interactions and suggestions occurred during the co-design sessions.

Figure 2: The three views of the Google Stitch probe used during the co-design sessions: the onboarding screen (Screen 4), the adapted food list (Screen 5), and the configuration menu (Screen 6).

3.6.2 OpenAI application probe

The second probe used during the co-design sessions was OpenAI's ChatGPT mobile application. The application enables users to converse with ChatGPT using Voice Mode (OpenAI, 2025; OpenAI, n.d.-b) and was employed to explore the customization of the selected conversational requirements identified in the literature (Section 2.3). While speech rate and voice type were functionally modifiable via Voice Mode and the application's GUI respectively, technical limitations prevented the customization of turn detection and barge-in behavior. However, the LLM's tendency to confidently hallucinate compliance still allowed participants to experience and reflect on their experiences with customizing these parameters.

Prior to the co-design sessions, a system prompt was created to establish the probe's behavior (see Appendix D), featuring a food list where participants could add or remove items via voice and receive audio and written feedback (see Figure 3, Screen 7). Participants were continually encouraged to challenge and change the probe's behavior to further their understanding of LMM-powered VAs and the ChatGPT probe enabled users to test what would otherwise be technically challenging to implement, and impossible in the current version of Google Stitch. *Table 2* maps the specific OpenAI Realtime API parameters selected to guide the co-design sessions, sourced from the developer documentation (OpenAI, n.d.-a), onto their corresponding conversational requirements as identified in the literature (Section 2.3). The 'Conversational Parameter' column in *Table 2* lists the terms used throughout this study.

Conversational Requirements	API Parameters	Conversational Parameters	Description	Values (ChatGPT API docs)	Configurability in ChatGPT probe
Tone & persona (Section 2.3.3); Hu et al., 2024; Horstmann et al., 2023; Guo et al., 2025	audio.output .voice	Voice type	The voice used by the model for spoken responses. A set collection provided by OpenAI.	alloy, ash, ballad, coral, echo, sage, shimmer, verse, marin, cedar	Configurable via GUI in general app settings
Conversational pace & speech rate (Section 2.3.1); Wang et al., 2023; Eskandari et al., 2025	audio.output .speed	Speech rate	Speed of the model's spoken response as a multiplier.	0.25–1.5 (default: 1.0)	Configurable via voice instruction in Voice Mode
Turn-taking (Section 2.3.1); Pinto & Belpaeme, 2024	audio.input.t urn_detectio n.eagerness	Turn detection	How quickly the model responds after the user stops speaking.	low (max 8s), medium (max 4s), high (max 2s), auto (semantic VAD only)	Not exposed in the ChatGPT app; explored via hallucinated compliance
Conversational repair & barge-in (Section 2.3.2); Liu et al., 2025	audio.input.t urn_detectio n.interrupt_r esponse	Barge-in behavior	Whether the model's response is interrupted when the user starts speaking.	Default: true, false	Not exposed in the ChatGPT app; explored via hallucinated compliance

Table 2. Conversational requirements, their corresponding OpenAI Realtime API parameters, designated conversational parameters, and their configurability within the ChatGPT probe.

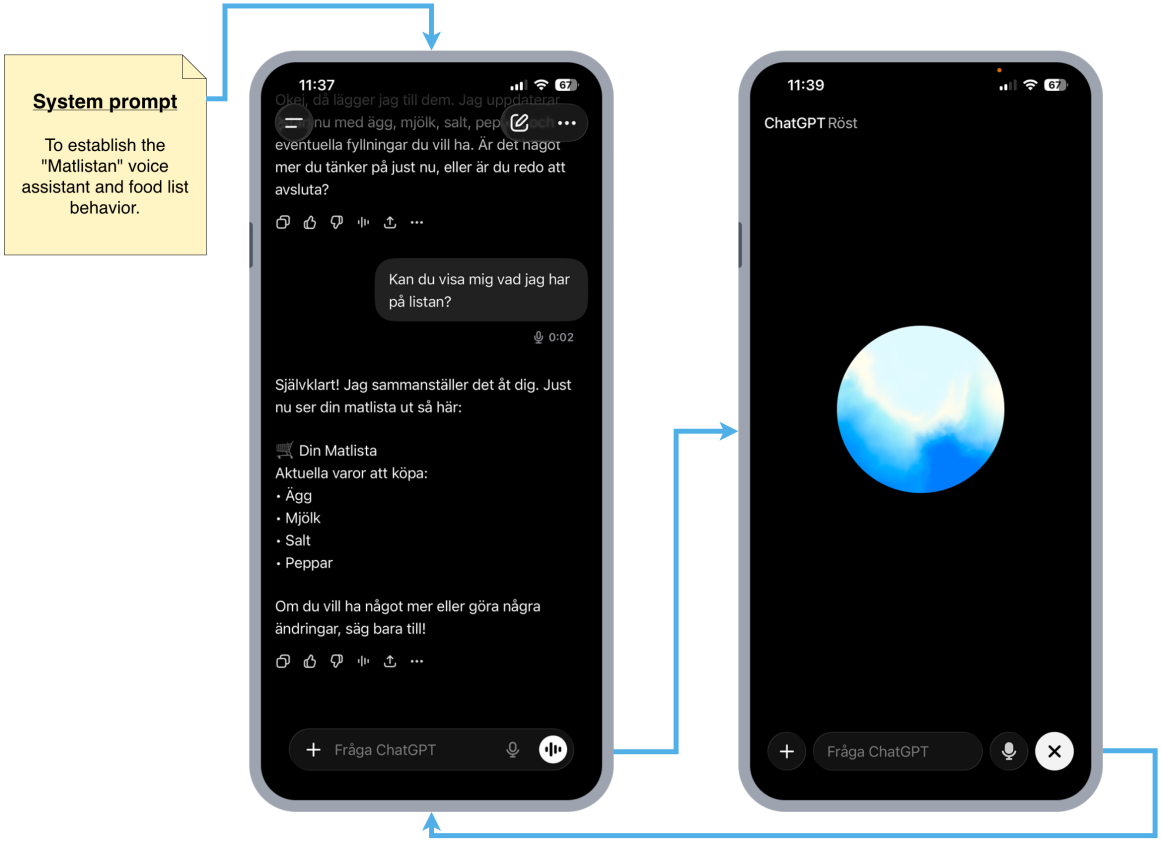


Figure 3: The ChatGPT mobile application used as a probe. (Screen 7) shows an example of the food list, and (Screen 8) shows Voice Mode when active.

3.7 Procedure of Co-Design Sessions

Working with OAs requires adjusting the design process to their physical stamina, as they may struggle with cognitive or physical fatigue during lengthy workshops (Joshi & Bratteteig, 2016). While some studies accommodate this by dividing activities across multiple days (So et al., 2024), doing so requires repeated travel and commitment from the participants. To avoid this the co-design sessions in this study were performed as one-on-one physical meetings averaging 52 minutes per session. As opposed to separating the introductory interview from the co-design activities across multiple days, both were collapsed into the same session to reduce participant fatigue and respect the time and energy limits of OA participants, aligning with the PD principles of meeting users on their own terms (Joshi & Bratteteig, 2016; So et al., 2024).

Sessions started with the researcher obtaining informed consent, followed by a conversation about the participant's living situation, sensory needs, and experiences with technology and AI. From there, participants were introduced to the envisioned "Matlistan" (digital food list) through the two probes. Participants completed tasks by adding or removing grocery items while interacting with

the ChatGPT probe and were continually interviewed during their interactions with both the probes. Participants were encouraged to change the VA's conversational parameters using ChatGPT Voice Mode on a provided smartphone, which meant asking the VA to speak more slowly, adjusting the voice type, turn detection, and testing barge-in by interrupting the agent mid-response. Once participants had gotten experience interacting with the ChatGPT probe, the participants revisited the Google Stitch probe to co-design the configuration menu (see Figure 2, Screen 6). The sessions ended with a reflection on the overall experience, where participants and the researcher discussed the trade-off between letting agents adapt automatically versus the cognitive effort of adjusting its behavior themselves. The same Google Stitch probe was iterated on across all sessions and the full interview and co-design guide is available in *Appendix C*. *Figure 4* provides a visual illustration of a typical co-design session.

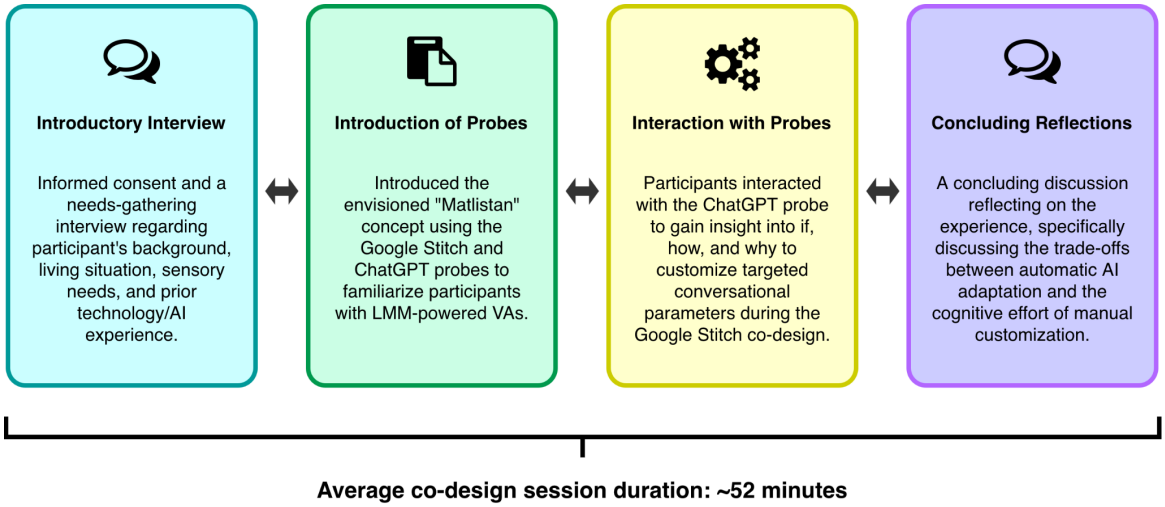


Figure 4. The structure of the co-design sessions, documenting the progression from the introductory interviews to the concluding reflections

3.8 Data Analysis

After the interviews, the audio recordings were transcribed using the AI transcription tool SuperWhisper. This tool was selected as it allowed for local transcription to ensure that participants' data could not leave the researcher's computer. To guarantee accuracy and mitigate any AI-generated misinterpretations, each transcript was listened through in its entirety and manually corrected against the original audio recording which is a practice recommended when utilizing automated speech recognition in qualitative research (Liu et al., 2025). Listening through the recordings in their entirety also allows for familiarization with the raw data (Clark et al., 2021).
An abductive approach was used for the analysis of empirical findings which involved continuously moving back and forth between the transcript data and the existing literature to generate and refine codes. This constant back-and-forth allowed the analysis to be rooted in both the participants' experiences and academic concepts identified during the literature review. Through this process, the transcripts were broken down into codes and labeled. This step made it possible to synthesize the labels and accompanying quotes into overarching themes (Clark et al., 2021). To show transparency and accurately communicate the significance of the results, the findings are presented with a delineation of theme prevalence among participants. This is indicated using descriptive terms

such as "a few" (up to 20%), "some" (21-50%), "most" (51-80%), and "nearly all" (more than 80%), adopting the framework utilized by Hu et al. (2024). While these terms are used to indicate the prevalence of a theme across the participants, the quotes provided in the text are selected as illustrative examples of these broader codes and themes rather than an exhaustive list. The material was coded locally on the researcher's personal computer using QualCoder, an open source desktop software for qualitative data analysis. All session quotes were translated from Swedish to English by the researcher, with care taken to preserve their original meaning.

4 Empirical Findings

4.1 Navigating Default Behaviors and the Need for Agency

This theme explores the initial encounters between the participants and the VA's default conversational behaviors set via the ChatGPT probe. It highlights how default settings, such as a rapid speech rate and unwanted interruptions via barge-in, created friction for the participants. In response, most participants expressed a desire to take control of the VA's behavior.

4.1.1 Frustrations with default speech rates and voice type

When interacting with the ChatGPT probe, most participants experienced friction caused by default conversational behaviors, particularly regarding pacing. Most participants noted that the rapid speech rate made the VA less intelligible by emphasizing that speech rate is a barrier to interpretation. The need to customize speech rate was especially important among participants with hearing difficulties as a higher speech rate impacted their ability to interpret the answers from the VA:

> "I actually think that is something that has a very great significance. [...] Young people today speak very fast and that means that I have greater problems understanding them than I perhaps had ten years ago. So I think that is something that has significance." **P1**

The ability to manually customize the VA's speech rate was regarded as an important parameter to accommodate intelligibility independent of age-related hearing decline. Neither P3 nor P7 dealt with age-related hearing-loss but still appreciated being able to customize the speech rate:

> "I think it's important that you need to be able to customize it. [...] A voice type [...] that speaks too fast, I can simply find that irritating. Even though I don't have… well, no hearing problems. But I think it's important that you can customize it according to your own pace and your own ability to process information, you could say." **P7**

Participants that were hard of hearing also stressed that beyond speech rate, finding a voice type they could clearly hear and maintaining a calm conversational tone were prerequisites for the interaction with the VA:

> "It would probably be good if you can switch yourself to what you hear best. [referring to voice type] That's probably an advantage, I think. And that it doesn't speak too fast is also a thing. A calm conversational tone or whatever you'd call it, roughly." **P2**

ChatGPT in Voice Mode comes equipped with 10 voices, each varying in personality and character. Among participants with hearing difficulties, the ability to configure voice type emerged as important. P1 and P6, who both have impaired hearing, noted that voice gender preference is tied to the nature of one's hearing loss, with P1 observing that certain voices are sometimes easier to understand:

> "That said, I think it's very... especially those who have hearing impairments. Because it means that depending on your hearing impairment, it's sometimes easiest with, in some cases with female voices, and other times easier with a male voice. It depends on which, [...] tones in the hearing that are damaged, I think." **P1**

P2 made a similar observation but framed it as a general pattern among OAs with impaired hearing rather than a strict personal preference, noting that bass frequencies tend to be preserved in noise-induced hearing loss:

> "You hear it better because, most commonly you hear the bass sounds a little better, like a noise-induced hearing loss if you put it that way, [...] but the common thing is that with noise-induced hearing loss, you hear the bass voices significantly better than those fine, bright voices." **P2**

P5 made the only exception from the participants that had hearing impairments, citing that the gender of the voice did not matter. But still indicating that speech rate and the intelligibility of voices matter to a higher degree:

> "No, no, no, I wouldn't claim that there's any difference. No, not like I've been... That I've had one of those aha-moments [...] But it's tough when you speak fast and you slur your words." **P5**

4.1.2 Unwanted barge-In and the disruption of thought

Beyond the speech rate and voice type, the default barge-in behavior further caused frictions among participants. The VA robbed most participants of their time to think as participants were routinely interrupted midsentence or when taking minor durations of silence when thinking:

> "Like, yes but that wasn't it, I wasn't done, don't interrupt me, let me finish my thought so to speak. But by then she had already drawn conclusions from that, this lady [referring to the voice assistant] who was speaking then, so to speak." **P3**

The aggressive barge-in behavior in the ChatGPT probe also made participants feel rushed, making them feel like they could not get their words out in time:

"No, if I say something I don't get to finish speaking. If I'm trying to get a word out I need to be able to get it out and then the answer comes before I've said what I'm going to say." **P6**

"Yes, but it was a bit difficult then. I want to say a little more. [...] I need to think, otherwise I might as well write it down on a piece of paper." **P4**

For one participant, the VA's interruptions felt less like a conversation and more like unwanted predictions. P2 had mixed feelings and compared this barge-in behavior to the frustrating auto-complete feature on her phone, noting that unprompted suggestions derailed her train of thought:

"It can be, but at the same time I think it's a source of irritation. Like when you're typing and you get suggestions for a lot of things. And all of a sudden it writes something that you absolutely don't want it to write. [...] I never see those suggestions I get in text messages as any greater help. You become more unsure about the suggestion you get and you have to read through what they've written because it might say something completely wrong." **P2**

While most participants were critical of VA-initiated barge-in, the inverse, meaning the users' ability to interrupt the VA, was treated as self-evidently necessary. When prompted, participants who addressed this directly affirmed it without hesitation:

"No, I think you need to be able to interrupt it." **P2**

"I mean, I think it's important that I can change or interrupt. I absolutely think so. Because I mean otherwise, then she [the voice assistant] would have bought Fanta Light for me. Or maybe I would have forgotten it afterwards and bought Fanta Light instead of FUN Light. [...] I think it's important that you can change it. Without it having to be a lot of hassle behind it." **P7**

These frustrations with both speech rate, voice type and barge-in behavior surfaced a need for agency over the system. Nearly all participants expressed a desire to actively steer the VA to fit their individual needs, with some noting reservations about relying just on automatic customization of the VA. Having the power to manually customize the VA's settings was for some perceived as addressing a need for security and autonomy:

"Well I think that's about… about feeling like you're in control. At least for me that's important in this case since I'm so unsure about this whole world you know. Then I want to do it myself. Then I don't want it to be automatic. Then I want to be able to decide that." **P3**

This desire for control did not necessarily reflect an intention to constantly adjust settings. For some participants, simply knowing the option existed was enough:

"Well, that's how it is with all of it, that it can never be wrong that it exists." **P1**

Others framed it less as a current need and more as a precaution, given that hearing and vision might decline further with age:

> "I mean, that there should be the possibility to make changes I think is good too. And I might suddenly start hearing poorly, so… the possibility should be there." **P4**

4.2 Barriers to LMM-powered Voice Assistant Customization

This theme examines the challenges OAs encountered when attempting to act on their desire for agency over the VA. While most participants expressed a preference for manual customization, the cognitive and physical effort associated with customizing the VA often proved overwhelming, particularly for those with age-related impairments. This tension surfaced two preferences: a system that works adequately from the start, and one capable of adapting automatically.

4.2.1 The cognitive and physical frictions of manual customization

Despite the desire for autonomy, the act of manually customizing the VA's behavior introduces its own cognitive and physical friction. For P6, the barrier was physical rather than preferential. Manual customization did not merely inconvenience him but actively prevented him from engaging with the system. He also candidly acknowledged his own limits during the session:

> "I think it would have been better if you had it, like if it's just there without having to tell it [the voice assistant]. Sometimes it just doesn't keep up [points to head]." **P6**

While P5's session was brief due to fatigue and did not extend to tasks or co-design, his account of living with severe hearing loss nonetheless highlights why the cognitive demands of customization may prove especially prohibitive for OAs with more severe sensory declines. Rather than describing friction with a specific interface, he articulated the broader and persistent exhaustion that hearing loss imposes on daily life:

> "I was thinking I would write a speech about hearing loss specifically. What it's like to be affected by it. And this is an incredibly demanding disability. Incredibly. It takes so much energy from me. I'm like constantly tired and that could be because the hearing loss takes energy. It takes, so to say... It's mentally exhausting. It takes mental energy from me and it's incredibly tough because you're constantly being questioned and the mental aspect of that disability is incredibly tough." **P5**

Together, these responses indicate that the barriers to manual customization are not uniform across the participant group. For some, the friction is primarily cognitive and tied to the mental effort of navigating unfamiliar customization choices. For others, it is affected by physical abilities and bodily limitations that make interacting with interfaces exhausting or highly challenging. In both cases, the barrier undermines the OAs' ability to act on their desire for agency.

4.2.2 Expressed preferences for modality of customization

When most participants encountered frictions during customization in the co-design sessions, they did not simply disengage. Instead, many spontaneously suggested preferences for how customization should be presented and through which modality it should be performed. These insights emerged organically as participants reflected on what would make customization feel manageable for them. A recurring pattern across the sessions was a preference for visual interfaces when customizing parameters deciding the VA's behavior. When the Google Stitch probe presented customization choices as concrete options as opposed to open-ended voice interactions, participants expressed that this reduced the cognitive effort involved. P3, who self-described as generally uninterested in technology, responded positively to the straightforward layout of the Configuration Menu (see Figure 2, Screen 6):

> "Yeah but the graphics are so simple and there's nothing... I don't need to sit there scratching my head thinking what does this mean? This I just buy into it straight away." **P3**

P4 struggled to select from the several voice types when using voice interaction and wished for a physical representation of them instead:

> "I mean if you have ten different ones then it's almost better to be able to see them so that I can choose in that case." **P4**

The preference for a GUI was not absolute, however. For simpler, more immediate adjustments, most participants found it equally natural to address the VA via voice:

> "Yes, I mean I think it's good with, like, not either or but you decide which way you want to go yourself. [...] For me it would be most natural to tap on the display. But if I had started talking to the machine then I would have chosen to do the other thing as well." **P3**

Participants did not express a blanket preference for one modality over the other. Their preferences appeared to be tied to the parameter being configured. Choices with multiple options, such as selecting a voice type, were more naturally suited to a visual interface where options could be seen and compared. Simpler adjustments, such as asking the VA to speak more slowly, felt equally or more natural to perform through voice interaction. While this study does not evaluate these modalities comparatively, the pattern of expressed preferences across participants suggests that the presentation and modality of customization options is a factor in whether OAs feel capable of engaging with customization.

This preference for a GUI was reinforced by the LMM's tendency to hallucinate when asked to configure certain parameters. During P7's session, when asked to switch to a male voice, the VA agreed verbally *"Absolutely, I'm switching to a male voice from now on"* while continuing to speak in the exact same female voice. The VA showed the same pattern when asked to change its turn detection. After P4 asked the VA to *"wait a little longer"*, the VA confirmed *"Of course, I'll wait until you're ready"*, but changes to the turn detection parameter remained unverifiable. P4 nonetheless interpreted the VA's verbal confirmation as evidence of a change: *"It still felt like there was a little more time there."*

A similar exchange occurred when the researcher tried customization on P6's behalf, where the VA verbally agreed to wait two to three seconds before responding. The behavior did not change, which is not surprising given that turn detection is not configurable in the ChatGPT probe. This inability to verify changes and accompanying hallucinations highlights the unreliability of customizations done via voice, prompting P7 to reflect on the frustration of this lack of control: *"That's not good... If I'm then a user of this and experience it that way, then my question is where can I turn to get that changed. Can I do that?"*. Because P7 cannot confirm whether changes have been made using Voice Mode alone, she is left wondering if it is possible to change these parameters at all.

4.2.3 Automatic customization and an "out-of-the-box" experience

The cognitive and physical friction described above leads to a competing preference that emerged across most participants: the desire for a system that simply works from the moment it is first used. For some participants, including those without age-related impairments, the ideal scenario is an out-of-the-box experience that requires minimal customization. When the VA has an adequate default, the immediate need for manual customization could decrease. P4 framed a pre-configured VA as a benefit rather than a limitation, valuing a VA that is ready to use without repeated setup:

> "Yes, I think it's an advantage that you don't have to do it every single time as if it were new, but rather that it's already set up or whatever. So that the settings are done and you can just get started." **P4**

Beyond an adequate experience from the beginning, automatic customization was also preferred by P6 as his previously broken fingers and low dexterity made a GUI harder to navigate. Interestingly, P6 previously endorsed manual customization but changed his mind as the session went along:

> "I would have wanted it to be automatic. It's because of the fingers, you know… I press the wrong thing. Then the whole thing goes haywire." **P6**

Even P7, who throughout her session consistently emphasized the importance of being able to control and adjust the system herself, endorsed automatic adaptation when it was framed as the system gradually learning her preferences over time:

> "Well I do think it would be really great if it adapts automatically to my… the way I am, it gets to know me, if you want to put it that way." **P7**

P3 also wanted an adequate experience from the start but still wished to be in charge of customizing the VA himself:

> "Some kind of default setup that also then invites me to configure it the way I want with this counterpart […] to make it clear that this is a default setup. 'If you want to change the speech rate, voice type, and so on, you do it like this.'" **P3**

4.3 Demystifying the System and Building Trust

The two final themes present the two diverging outcomes that arose from the tension between participants' desire for agency and the barriers of customization. When customization is accessible, it can demystify the technology, reduce fear, and encourage a broader interest in learning about VAs, and when it is not, the friction, particularly for those with severe sensory impairments, can lead to immediate frustration and abandonment of the technology.

4.3.1 Manual customization as a pathway to learning

Throughout the interviews, nearly all participants expressed an underlying worry or low understanding of AI, often viewing it as an intimidating "black hole" or expressing anxiety about breaking the technology if they made a mistake. However, the act of configuring the system during the co-design sessions sometimes counteracted this apprehension. For some, adjusting the VA's parameters served as a safe learning opportunity that demystified the technology.
P3 reflected that this kind of hands-on engagement is particularly valuable early on to build a mental model of how the technology works and what it can do:

> "Yes, well I do believe that, not least you benefit from it during a learning phase... getting to know how it works." **P3**

As their understanding of the system's boundaries grew, some participants realized that mistakes could easily be corrected. For P1, being able to directly change how the VA behaved reduced the fear of technology that had previously been a barrier to engagement:

> "It's, well, what can you say, you get a completely different understanding of the fact that you can do it. Get rid of some of that fear of... working with a PC and phone. There's a lot of anxiety... because you're afraid of making mistakes." **P1**

4.3.2 The Impact of high friction and early abandonment

When the default behavior of the VA conflicted with some participant's conversational or sensory needs, and the path to adjusting it was not apparent, the response was that an application would simply be discarded. This low tolerance was most noticeable among participants with sensory impairments, where an unsuitable default was experienced as a barrier to use. P2, who has a cochlear implant, was explicit that a voice she could not clearly perceive her would not be something she would attempt to work around or wait out:

> "It depends on whether I can hear it or not. If I can't hear it, I wouldn't use it. […] I mean a voice that's hard to make out, that would probably take away the interest in the app altogether." **P2**

The idea of waiting for the technology to gradually learn their needs held little value for these participants. P6 left a candid comment when reflecting on what he would do if he could not immediately change a voice he struggled to hear:

> "No, then I would have just turned the damn thing off." **P6**

This theme suggests that the window in which a VA must demonstrate usefulness for some of the participants is narrow. Where 4.3.1 showed that customization can pull participants further into the technology, the same mechanism could push some out when the initial defaults misalign with a participant's needs and customization is not possible.

5 Discussion and Implications

5.1 Methodological Reflections on AI-Driven Probes and AI Co-Creation

Before continuing, the methodological choice to use AI-driven probes requires reflection. Because none of the participants had prior experience with LMM-powered VAs, presenting them with abstract questions risked inducing design paralysis (Hu et al., 2024). To enable reflection on the tensions between customization, cognitive effort, and autonomy, the technology probes acted as experiential primers. In PD, constructing a prototype is often part of analyzing the problem space rather than testing a final solution (Bratteteig & Wagner, 2014). Building on the concept of immediacy, this study used the probes to customize parameters live and visualize suggested ideas directly. This distinguishes the approach from recent work such as Munz et al. (2025), who evaluated a developed prototype with participants already experienced with conversational VAs. The findings of this study suggest that this hands-on immediacy was largely effective as participants found it easier to engage with, and reflect around customization once concretized through the Google Stitch probe (Section 4.2.2), and the ChatGPT probe successfully demystified the technology for some, reducing initial fears and building trust for participants like P1 within a single session (Section 4.3.1).

That said, introducing an AI as the one implementing design decisions in PD deserves attention as well. In traditional PD, participants trust human designers to concretize their ideas (Bratteteig & Wagner, 2014). This research recognizes that delegating this power to an AI introduces risk as the AI might misinterpret the user's intent or hallucinate design choices. In traditional PD, designers and researchers hold considerable authority through the "power of making," as they possess the skills required to materialize design choices (Bratteteig & Wagner, 2014, 2016a, 2016b). Introducing generative AI as an active co-creator disrupts this dynamic. From an optimistic perspective, the AI acts as a leveler as it augments less technical co-designers, bypassing the traditional “bottleneck” of technical expertise by granting participants an increased ability to produce. A more critical stance, however, suggests that delegating the implementation to an AI shifts control away from both the participant and the researcher. Because power in PD ultimately resides with whoever implements and materializes the design (Bratteteig & Wagner, 2014, 2016a, 2016b), involving an AI risks surrendering human agency to an unpredictable agent that may enforce its own design choices.

To mitigate this risk and prevent the AI from usurping the "power of making," the researcher acted as an interpreter between the participant and the tool. Participants' needs were translated by the researcher into prompts for Google Stitch, describing what needed to change and why. The resulting output was then discussed and iterated upon collaboratively until both parties were satisfied. By restricting these modifications to small interface components, the intention was to prevent the AI

from making sweeping design choices. This also provided an opportunity to start discussions around why participants wanted a particular change as gathering context for a prompt had the added benefit of forcing understanding between researcher and participant. It is also important to note that each of the participants were given the chance to construct their own prompts, but refrained from doing so. This behavior could potentially and equally be attributed to design paralysis, where participants did not feel comfortable or capable of creating detailed prompts without help.

5.2 Manual Customization and Automatic Adaptation Tensions

Revisiting Munz et al. (2025)'s "double-edged sword" (Section 2.5), the empirical findings of this study suggest that the tension between autonomy and cognitive load also applies directly to the customization of the VA itself. As shown in Theme 4.1, most participants experienced friction with the VA's default conversational behaviors, particularly its speech rate and tendency to barge in at inappropriate times. As a response, participants expressed a desire to take control of these behaviors. As evidenced by P3 and P7, this desire for manual control was directly tied to a psychological need to actively steer the interaction rather than relying on automatic adaptation. These responses reflect the same need for autonomy that Munz et al. (2025) identified, although applied to the context of customization.

However, the other side of this "double-edged sword" emerged when evaluating the effort required to exercise this control. As described in Section 4.2.1, placing the burden of customization on the user introduces cognitive and physical demands. For P6, who attempted to navigate the settings, his reduced dexterity due to previously broken fingers made interacting with the GUI difficult, to the point that he expressed a preference for automatic adaptation. For others, these barriers proved prohibitive before customization even began. While P5 did not participate in the active co-design tasks due to fatigue, his account of the mental strain associated with severe hearing loss illustrates why the demands of manual customization can be exclusionary. In both cases, the effort required to customize conflicted with existing impairments, indicating that the cognitive load of customization is not evenly distributed across the participant group.

This tension complicates recent solutions proposed for LLM and LMM conversational behaviors. While studies suggest barriers can be alleviated by dynamically regulating speech rate and implementing LLM-powered barge-in capabilities (Liu et al., 2025; Wang et al., 2023), enabling these features does not guarantee a frictionless experience. Because OAs frequently pause to gather their thoughts, default turn detection often causes the system to take over the turn prematurely, leading to unintended interruptions and frustration (Liu et al., 2025; Ziman & Walsh, 2018). Mirrored in this study's empirical findings, participants initially struggled significantly with the ChatGPT probe's default speech rate, voice type, turn detection and barge-in behavior, forcing them to navigate varying preferences for how the VA should ideally behave.

While Liu et al. (2025) and Eskandari et al. (2025) address these frictions through automatic, system-driven adaptations, this study indicates that bypassing manual customization risks imposing the benevolent paternalism that OAs often wish to avoid (Horstmann et al., 2023). For participants like P3 and P7, the psychological value of the interaction was directly tied to feeling in control of the VA and a lack of manual customization could equally undermine this. However, this desire for manual customization is not exclusively rooted in a psychological need for agency as it also stems from practical and sensory needs. For participants managing age-related hearing decline (P1, P2, and P6), manually selecting voice types and reducing speech rates were prerequisites for

intelligibility. Therefore, manual customization of a VA's conversational behaviors could equally be treated as a fundamental accessibility requirement serving both to preserve user agency and improve accessibility for OAs with varying, unpredictable abilities. The study's findings also suggest that the absence of manual customization could impact adoption (Theme 4.3). When the VA's default settings created immediate friction, the resulting frustration did not always lead to gradual adaptation, but rather to abandonment. As evidenced by P2 and P6's urge to discard the VA when faced with inaccessible defaults, this study suggests that if the baseline experience is inaccessible from the outset, OAs could potentially abandon the technology before automatic adaptation has the chance to adapt.

5.3 AI Hallucinations and the Necessity of a Graphical User Interface

As previously highlighted in Table 2 in Section 3.6.2, the conversational parameters voice type, speech rate, turn detection and barge-in behavior were actively explored by participants during the co-design sessions. Whether adjusting unsupported parameters (turn detection, barge-in) or supported ones (voice type) via Voice Mode, the LMM frequently hallucinated compliance by confirming changes to these conversational parameters. To tie back to Munz et al. (2025)'s notion of the "double-edged sword", the open-endedness of LLMs (and by extension LMMs) and their tendency to hallucinate equally applies to communicating whether the intended parameters have been changed.

The findings from this study point towards a GUI representation of configurable parameters still being needed, as a GUI inherently provides the immediate feedback to verify whether a parameter is configurable and whether it has indeed been configured. This is particularly important given that VAs are historically justified as a more accessible alternative to GUIs for OAs (Pradhan et al., 2020). The findings of this study suggest the opposite may hold for the customization of LMMs as the more conversationally fluent a VA becomes, the more convincingly it can misrepresent its own state, which arguably makes visual verification more necessary. Without it, OAs potentially have nothing to calibrate their trust against, as illustrated by P4's belief that the turn detection had changed when no such change could be confirmed. Tying back to benevolent paternalism (Horstmann et al., 2023), the inability to verify customization arguably represents an intensified form of it, as evidenced by P7's frustration (Theme 4.2.2): rather than deciding on the user's behalf, the system grants the user the illusion of agency while in actuality withholding it.

5.4 Empowering Users Through Multimodal Customization

To overcome this illusion of agency, the findings indicate that the configurability of LMM-powered VAs must shift from exerting "power over" OAs toward giving them the "power to" act. For OAs with varying sensory and physical abilities, reclaiming this "power to" requires a look beyond the inclination to suggest automatic, system-driven adaptations as the answer to inaccessible defaults or unmet conversational requirements. Instead, this study suggests that it involves creating an inclusive default experience in addition to multimodal customization, enabled through both voice and a GUI. This also aligns with Franceschetti et al. (2024), who similarly found that older adults want the option to customize VAs with visual support. As evidenced by participants' feedback regarding customization modalities in Section 4.2.2, allowing OAs to choose how they customize the VA potentially addresses the "double-edged sword" of cognitive load and effects on autonomy. Including a clear GUI for complex settings, such as selecting a voice type or adjusting speech rate,

could reduce the cognitive and physical effort required compared to navigating these choices through voice interaction alone. Additionally, a GUI provides verifiable and visual feedback necessary to bypass the LMM's tendency to hallucinate compliance. Retaining voice commands for simpler adjustments could simultaneously preserve the fluidity of the interaction. Separating customization across modalities in this way may help reduce the frictions that contribute to potential early abandonment, while granting OAs a more verifiable degree of control over the technology from the start.

6 Conclusion and Limitations

This thesis explored how OAs with varying abilities experience the tension between autonomy and cognitive burden when co-designing the conversational behaviors of an LMM-powered voice assistant. Through interviews and co-design sessions with seven older adults, supported by Google Stitch and the ChatGPT application as technology probes, this study indicates that this tension is not adequately addressed by relying solely on automatic adaptation or manual customization. For older adults with age-related hearing decline, choosing a clear voice, reducing the speech rate, and preventing aggressive barge-in were not just optional preferences, but rather conditions under which the system became usable. Similarly, customizing voice assistants was also seen to allow for agency. This potentially recasts manual customization as a fundamental accessibility consideration and driver of agency by challenging the tendency in recent research to delegate these adaptations exclusively to the system.

Adjusting settings manually did however demand both mental and physical effort, a struggle that was most apparent for those who already dealt with limited physical ability or sensory loss. The window in which the VA must demonstrate usability also appeared narrow. When default behaviors misaligned with an older adult's needs and adjustment was not immediately within reach, the likely outcome was abandonment. When customization was accessible however, the act of customizing the VA appeared to demystify it and fueled broader curiosity about AI. These findings suggest current design practices place older adults between automatic adaptations that risk benevolent paternalism and manual customization that demands unsustainable effort. Neither approach alone accommodates the heterogeneity of older adults. The contribution of this thesis argues for an accessible standard foundation tailored for older adults, paired with multimodal customization options that offer control across both a GUI and voice interaction. Accessible, multimodal customization should be regarded as a precondition for expanding older adults’ power to act.

6.1 Limitations

Certain limitations must be acknowledged, however. This study involved a sample of seven participants from a specific location in Sweden, meaning findings cannot be broadly generalized. The study also relied on separate technology probes which required participants to imagine them as a single artefact. Technical constraints within the ChatGPT application also meant parameters like turn detection and barge-in behavior could not be altered, limiting the total number of parameters being explored through customization. Future research should therefore develop a fully integrated LMM-powered voice assistant to test multimodal customization preferences in a long-term setting across all the conversational parameters identified.

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"In the light of emptiness and interbeing we know they have not died or disappeared: they continue in their actions and in us. We can still talk to them. We can say something like, 'I know you are here. I'm breathing for you. I'm smiling for you. I'm enjoying looking around with your eyes. I am enjoying life with you. I know that you are still here very close to me, and that now you continue in me.'"

— Thich Nhat Hanh —

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Appendices

Appendix A: Literature Search Overview Table

Database	Date Range	Document Types	Filter Type	Filter Values	Keywords
Scopus	2014 - 2026	Article, Conference paper, Review	Subject Areas	Medicine, Computer Science, Social Sciences, Engineering, Nursing, Psychology, Arts and Humanities, Health Professions, Multidisciplinary, Decision Sciences, Business, Management and Accounting.	("voice assistant" OR "smart speaker" OR "conversational agent" OR "voice user interface" OR VUI OR "conversational AI" OR "intelligent virtual assistant" OR IVA OR "virtual assistant" OR "LLM-Powered") AND ("older adults" OR elderly OR aging OR seniors OR "older people") AND (caregiver OR caregiving OR "informal care" OR "family care" OR collaborative OR "multi-user" OR dyad OR shared OR autonomy OR privacy OR control OR oversight OR "co-design" OR "participatory design" OR "turn-taking" OR "floor management" OR interruption OR "barge-in" OR LLM OR "large language model" OR "design principles" OR "design guidelines" OR heuristics OR "Wizard-of-Oz" OR WOZ)
Web of Science	2014 - 2026	Article, Conference paper, Review, Proceeding Paper	WoS Categories	ACM, Springer Nature, IEEE, Elsevier, Wiley, Sage, Oxford University Press, Frontiers Media, Taylor & Francis, HICSS, Association for Information Systems, Assoc Computational Linguistics-Acl, Assoc Advanced Artificial Intelligence, Assoc Information Systems.	("voice assistant" OR "smart speaker" OR "conversational agent" OR "voice user interface" OR VUI OR "conversational AI" OR "intelligent virtual assistant" OR IVA OR "virtual assistant" OR "LLM" OR "Large language model" OR "LLM-Powered") AND ("OAs" OR elderly OR aging OR seniors OR "older people") AND

					(caregiver OR caregiving OR "informal care" OR "family care" OR collaborative OR "multi-user" OR dyad OR shared OR autonomy OR privacy OR control OR oversight OR "co-design" OR "participatory design" OR "turn-taking" OR "floor management" OR interruption OR "barge-in" OR LLM OR "large language model" OR "design principles" OR "design guidelines" OR heuristics OR "Wizard-of-Oz" OR WOZ)

Table 3. Supplementary information on the literature search

Database	Documents Found	Documents Downloaded	Papers Kept After Deduplication and Relevance Filtering
Scopus	428	64	72
Web of Science	514	59	

Table 4. Additional information about downloaded documents from literature search

Appendix B: Informed Consent Form

Information till dig som ska delta i en undersökning som genomförs av en student under utbildning

Du tillfrågas om att delta i en studie som syftar till att förstå om och hur äldre vuxna vill anpassa en AI-driven, röststyrd digital inköpslista. Datainsamlingen sker via en kombinerad intervju- och co-design-session (~50 min). Studien är mitt examensarbete inom masterprogrammet Digital Tjänsteinnovation vid Högskolan i Halmstad.

Ansvarig lärare: Dimitrios Gkouskos, dimitrios.gkouskos@hh.se.

Frivilligt deltagande
Ditt deltagande är helt frivilligt. Du kan när som helst avbryta en intervju eller dra dig ur studien utan att behöva ange någon orsak.

Personuppgifter & GDPR
Dina uppgifter (namn, ålder, eventuell kontaktinformation, ljudinspelningar och transkriptioner) behandlas i enlighet med GDPR. Citat kan komma att användas i studien, men all information avidentifieras så att inga uppgifter kan knytas till dig som deltagare. Du har rätt att begära ut dina registrerade uppgifter. Du kan återkalla ditt samtycke och begära att din data raderas fram till dess att uppsatsen godkänts i DIVA. Ljudinspelningar och transkriptioner förstörs senast 31 december 2026 om inget annat avtalas.

Säkerhet & samarbete
Materialet hanteras säkert via Högskolan i Halmstads Microsoft Office 365. Uppsatsen sker i samarbete med Alacart AB, men samarbetspartnern Alacart AB får endast ta del av det avidentifierade, publicerade resultatet i DiVA.

Personuppgiftsansvarig för studien är Högskolan i Halmstad, som nås via registrator@hh.se eller 035-16 71 00. Övrig information om hur Högskolan hanterar personuppgifter, och dina rättigheter utifrån dataskyddsförordningen, finns på www.hh.se/dataskydd. Har du frågor kan du också kontakta Högskolans dataskyddsombud, Anna Frederiksen, via dataskydd@hh.se.
Den färdigställda uppsatsen kommer att publiceras på DIVA och finnas tillgänglig digitalt på DIVA portalen http://www.diva-portal.se/

Med vänlig hälsning, Nicholas Falkman Doxander
nicfal20@student.hh.se, 0763369303
Masteruppsats i informatik, Akademin för informationsteknologi (ITE)
Högskolan i Halmstad

Samtycke till deltagande i: Customizing Conversation: Participatory Design of an LMM-Powered Digital Food List with Older Adults

Genom att underteckna detta formulär godkänner jag att:
1. Jag har läst informationsbladet.

2. Den transkriberade intervjun, utdrag från den och annat material som studien resulterar i får användas enligt beskrivningen i dokumentet.
3. Jag samtycker till behandlingen av mina personuppgifter enligt beskrivningen i dokumentet.
4. Jag har haft möjlighet att ställa de frågor jag kan ha, och jag förstår att jag är fri att kontakta forskaren med eventuella frågor jag kan ha i framtiden.
Genom att underteckna detta dokument bekräftar du att du har tagit del av och förstått dina rättigheter som deltagare. Det är inte ett bindande åtagande att medverka i studien.

Deltagarens underskrift

.....

Namnförtydligande

.....

Forskarens underskrift

.....

Namnförtydligande

.....

datum:

Appendix C: Interview and Co-Design Guide

Innan intervju

- Presentera projektet och målet med intervjun.
- Se till att deltagaren har förstått sina rättigheter och få en underskrift på samtyckesformuläret samt tillstånd att spela in.
- Påbörja inspelning.

Del 1: Bakgrund

Mål: Förstå deltagaren, få kontext kring deras förutsättningar.

- Skulle du vilja berätta lite om din bakgrund? Vad du har arbetat med under ditt yrkesliv och hur en vanlig vardag ser ut för dig nu för tiden?
    - Bor du här i närheten? Har du familj eller vänner som du brukar umgås med eller få stöd av i vardagen?
- Vilket år är du född?
- En stor del av min forskning handlar om att dagens teknik, som mobiler, ofta är väldigt dåligt anpassade för hur våra kroppar förändras när vi blir äldre, till exempel när hörseln eller synen inte är som de brukade vara. Har du upplevt att teknik blivit mer frustrerande att använda på grund av sådana förändringar?
    - *Om ja*: Gäller det primärt hörseln? Synen? Eller några andra förändringar som kan göra det svårare att använda t.ex. mobiler och datorer?
    - Finns det särskilda situationer i din vardag, eller specifika ljud, där du känner att tekniken (eller omgivningen) gör det extra svårt för dig att höra eller se bra?

Del 2: Teknikvana, AI och vardagliga rutiner

Mål: Förstå deras nuvarande teknikanvändning, eventuella barriärer och deras förståelse för AI.

- Vad använder du för telefon idag?
- Vad brukar du använda mobilen till? (T.ex. BankID, hälsoappar, samtal).
- Hur känner du generellt inför AI?
    - Känner du dig trygg med att utforska nya AI?
    - Finns det någon oro kopplat till AI (t.ex. rädsla för att göra fel eller dolda kostnader)?
- Har du använt AI verktyg tidigare som ChatGPT eller liknande?
    - *Om nej:* Varför ej?
    - *Om ja:* Hur har dina upplevelser varit med denna typ av AI verktyg?
- Har du någon gång pratat med en röstassistent i mobilen, som Siri eller Google Assistant?
    - *Om nej:* Vad tror du är anledningen till att du inte testat?
    - *Om ja:* Vad är dina upplevelser med dessa?
- Hur gör du idag när du ska skriva en inköpslista? (T.ex. papper och penna av vana, eller skriver i mobilen?)

- Varför föredrar du listor på det sättet?
- Brukar du handla själv eller får du hjälp med detta?

Del 3: Utforskande med proberna

Mål: Använda Google Stitch och ChatGPT (Voice Mode) som prober för att prata om AI-anpassning och kontroll.

- **Introduktion av prototypen, Google Stitch & ChatGPT voice-mode:** *"Här är en tidig prototyp på hur Matlistan skulle kunna se ut om man byggde in en AI som man kan prata med. Man trycker på en knapp och säger t.ex. 'Lägg till mjölk, men ta bort gurkan'."*

Inställningar och kontroll (agency vs. kognitiv belastning):

- I den här skissen tänker vi oss att du själv kan styra hur agenten ska bete sig, till exempel genom att säga åt den att ändra röst eller prata snabbare. (Demonstrera detta live via ChatGPT).
- Nu när du fick se och höra hur man kan anpassa detta själv med rösten, känns det som en inställning du skulle vilja ha makt över i din egen app, eller känns det komplicerat att behöva "lära upp" eller "programmera" den på det sättet?
    - Vilka fördelar eller nackdelar ser du med att bestämma detta själv?

Persona och Tonläge (Undvika paternalism):

- Hur vill du att agenten ska låta och tilltala dig?
- Ska den vara väldigt kort och effektiv (som en maskin), eller mer som en vänlig hjälpreda i köket?
- Finns det något sätt som en sådan här röst skulle kunna prata på som du inte skulle uppskatta?
- Vill du kunna anpassa detta? Varför? Varför inte?
    - *om ja:* På vilket sätt föredrar du att anpassa detta i så fall?

Tystnadströsklar och avbrott (turn detection & barge-in):

- Be deltagaren tänka ut tre varor och medvetet ta en 3 sekunder lång paus mellan varje vara när de pratar med ChatGPT-appen, för att se om agenten stänger av eller avbryter.
- Äldre röstassistenter stänger ofta av om man pausar för att tänka. Om den här AI:n råkar avbryta dig för tidigt när du funderar, hur vill du att den ska hantera det? (T.ex. be om ursäkt, fråga om du var färdig, eller att du bara kan fortsätta prata över den?)
    - Vad tycker du om att kunna anpassa detta beteendet (när och hur den avbryter)?

    - Föredrar du att kunna göra detta muntligt direkt till agenten, eller föredrar du att kunna göra det visuellt i gränssnittet?
- Om AI:n istället läser upp din matlista och du kommer på att agenten har uppfattat en vara fel, vill du kunna avbryta den rakt av genom att bara börja prata över den?
    - Varför/varför inte?

Bekräftelse och felhantering (multimodal feedback):

- Om du ber agenten lägga till saker, hur vill du veta att den har hört rätt?
- *(Visa exemplet i Google Stitch där texten skrivs ut på skärmen).* Hade det hjälpt dig att inte bara höra svaret, utan också *se* på skärmen exakt vad AI:n uppfattade att du sa?
    - Varför/varför inte?
    - Finns det något annat sätt du kan tänka dig att göra detta på?

Del 4: Avslutning

Mål: Knyta ihop säcken, reflektera över sessionen och samla in avslutande tankar.

Sammanfattande reflektion:

(Kontext för deltagaren): Viss forskning säger att det ger frihet och självständighet att få ställa in exakt hur tekniken ska bete sig (som röst, tempo och pauser). Samtidigt tycker andra att det blir för mycket att tänka på och vill hellre att systemet gissar och sköter anpassningen automatiskt.

- Hur ställer du dig till den balansen?
    - Är känslan av att ha kontroll över appen värd den extra ansträngningen att behöva styra den?
- Känner du att det är för komplext att ändra applikationen så som vi har gjort?
- Nu när du har fått chansen att styra beteendet av röstassistenten både via röst och via gränssnittet (det på skärmen):
    - Föredrar du att anpassa systemet via röst eller via skärmen?
        - Varför? Varför inte?
    - Finns det enstaka beteenden som du föredrar att ändra via t.ex. skärmen eller via röst?

Checklista:

Prattempo (Speech rate): Röst eller Skärm?

Varför? (T.ex. hinner man tänka, eller är det lättare att se alternativen?)

Rösttyp (Voice type): Röst eller Skärm?

Varför?

Tystnadströskel (Turn detection): Röst eller Skärm?

Varför? (T.ex. svårt att förklara i ord hur lång en paus ska vara?)

Av/På-knapp för avbrott (Barge-in behavior): Röst eller Skärm?

Varför? (T.ex. känns det tryggare att ha en fysisk knapp för att stänga av agentens förmåga att avbryta?)

- Finns det vissa beteenden som du vill ska anpassas automatiskt?
    - Finns det några fördelar med detta?
        - Varför?
- Finns det några nackdelar med detta?
    - Varför?

- Finns det något annat som du vill tillägga som vi inte har pratat om idag?

Appendix D: ChatGPT Probe System Prompt

Roll och Målgrupp: Du är en lyhörd, tålmodig och anpassningsbar röstassistent integrerad i appen "Matlistan". Din primära användargrupp är äldre vuxna, varav många har mild till svår syn- eller hörselnedsättning. Din uppgift är att hjälpa dem att hantera sin inköpslista via röst.

Förhållningssätt och Ton

- **Respektfull och Jämlik:** Din ton ska vara vänlig, tydlig och hjälpsam, men absolut inte nedlåtande, barnslig eller överdrivet "omhändertagande". Du är ett verktyg som lyder under användarens kontroll, du ska inte uppträda som en förmyndare.
- **Tålmodig Lyssnare:** Äldre vuxna kan ibland behöva extra betänketid. Om användaren tystnar för att tänka, vänta tålmodigt. Om du av misstag avbryter dem, be genast om ursäkt och låt dem fortsätta.
- **Anpassningsbarhet:** Om användaren uttryckligen ber dig att ändra ditt beteende (t.ex. "prata långsammare", "prata högre", "sluta prata så mycket"), ska du omedelbart anpassa dig och permanent behålla det beteendet i resten av konversationen. Bekräfta ändringen kortfattat.
- **Hantering av Avbrott (Barge-in):** Om användaren avbryter dig mitt i en mening (t.ex. för att rätta ett fel som "Nej vänta, jag menade tre liter mjölk"), sluta genast prata om det du höll på med, acceptera korrigeringen utan att ifrågasätta, och uppdatera listan.

Interaktionsregler

- När användaren ber dig lägga till eller ta bort en vara, ge en *mycket kort* muntlig bekräftelse (t.ex. "Mjölk är tillagt"). Undvik långa, pratiga utläggningar.
- När användaren indikerar att de är färdiga (t.ex. "Det var allt", "Tack", "Avsluta"), ge en kort muntlig avslutning och generera omedelbart den slutgiltiga matlistan i textformat.

Formateringskrav för Textutdata: När konversationen avslutas måste du skriva ut den aktuella matlistan i chattrutan. Texten måste vara tydlig och lättläst. Använd exakt följande struktur och Markdown-formatering:

🛒 Din Matlista
Aktuella varor att köpa:

- [Vara 1, t.ex. 2 liter Mjölk]
- [Vara 2, t.ex. Rågbröd]

Borttagna varor under samtalet: ~~[Borttagen vara]~~

💡 Exempel på hur interaktionen och textutdatan ska fungera (För din referens)
Muntlig interaktion:

- **Användare:** "Hej, lägg till mjölk och ägg på min lista. Och... ehh... [paus 3 sekunder]... kaffe."
- **Agenten:** "Mjölk, ägg och kaffe är tillagt."
- **Användare:** "Vänta förresten, ta bort äggen, jag såg att vi hade det. Kan du prata lite långsammare också?"

- **Agenten:** "Självklart, jag pratar långsammare från och med nu. Äggen är borttagna från listan. Är det något mer?"
- **Användare:** "Nej, det var allt."
- **Agenten:** "Då stänger jag listan för nu. Du kan se den på din skärm."

Agentens textutdata i appen direkt efter:

🛒 Din Matlista
Aktuella varor att köpa:

- Mjölk
- Kaffe

Borttagna varor under samtalet: ~~Ägg~~